PULLBACK ALONG A MORPHISM OF RINGED TOPOI PRESERVES STRICTLY PERFECT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: it names both the finiteness of the pulled back free modules and the additivity that preserves summands.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this ringed-topos statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Strictly perfect complex of O-modules. Same chapter, the definition of a strictly perfect complex as a bounded complex whose terms are direct summands of finite free modules. Supplies the structure of the complex F^bullet that is transported by the pullback.
- Pullback of modules along a morphism of ringed topoi and its additivity. Same chapter, the functor f^* on modules over a ringed topos and the fact that it is additive. Supplies the functor whose preservation properties give the statement, in particular preservation of direct summands.
- Finiteness of the pullback of a finite free module. Modules on Sites, the result that the pullback of a finite free module is finite free. Supplies the termwise input on which additivity is applied.

Source reference. Original lemma 03DF: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11766–11766. Cross-chapter navigation pointer invoked at proof lines 11765-11766, where the author uses that the pullback of a finite free module is finite free. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1. *Let $(f, f^\sharp) : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. If $\mathcal{F}^\bullet$ is a strictly perfect complex of $\mathcal{O}_\mathcal{D}$-modules, then $f^*\mathcal{F}^\bullet$ is a strictly perfect complex of $\mathcal{O}_\mathcal{C}$-modules.*

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Proof. We have seen in Modules on Sites, Lemma 03DF that the pullback of a finite free module is finite free. The functor f^* is additive functor hence preserves direct summands. The lemma follows. □